ASSOCIATION RULE MINING

Apriori Algorithm

From transactions to meaningful association rules

Definitions • One 8-transaction example • Manual calculations • Interpretation • Python notebook

Same products and calculations on every slide

TRANSACTIONS

Milk + Bread + Butter

Milk + Bread + Eggs

Bread + Butter

LEARNING PATH

One example, six logical steps

We will calculate everything using the same 8 supermarket transactions.

01

Transaction data

8 bills and 6 products

02

$C_1 \rightarrow L_1$

Count and discard products

03

$C_2 \rightarrow L_2$

Count and discard pairs

04

$C_3 \rightarrow$ Stop

Check subsets and prune

05

Association rules

Calculate support and confidence

06

Lift + Python

Interpret and implement

DEFINITION

What is Apriori?

An algorithm for finding item combinations that occur frequently in transaction data.

APRIORI — CORE PURPOSE

Discover frequent itemsets by counting support and eliminating combinations that cannot meet the minimum support threshold.

Example

Milk + Bread

Bread + Butter

- Apriori finds frequent combinations first.
- Association rules (A → B) are evaluated afterward.

KEY IDEA Frequent together does not automatically mean a useful recommendation.

Our supermarket transaction data

Each row is one basket. We will not change the dataset during the lesson.

Transaction	Products purchased
T1	Milk, Bread, Butter
T2	Milk, Bread, Eggs
T3	Milk, Bread, Butter, Cheese
T4	Bread, Butter, Eggs
T5	Milk, Bread, Butter
T6	Milk, Bread, Juice
T7	Bread, Butter, Cheese
T8	Milk, Eggs

TOTAL BASKETS

8

T1 through T8

UNIQUE PRODUCTS

6

Milk, Bread, Butter, Eggs, Cheese, Juice

Minimum support 50%

KEY DEFINITIONS

Support and minimum support

Minimum support controls which combinations continue to the next iteration.

Support(X) = Transactions containing X ÷ Total transactions

MINIMUM SUPPORT

50%

Chosen threshold

8 transactions × 50% = 4 transactions

Any itemset needs a transaction count of at least 4.

KEY IDEA Minimum support is tested at every itemset size: 1 item, 2 items, 3 items...

STEP 1 • C_1

Count all individual products

C_1 means candidate 1-itemsets: six products before filtering.

Product	Count	Support
Milk	6	75%
Bread	7	87.5%
Butter	5	62.5%
Eggs	3	37.5%
Cheese	2	25%
Juice	1	12.5%

REFERENCE — 8 TRANSACTIONS

ID	Products
T1	Milk, Bread, Butter
T2	Milk, Bread, Eggs
T3	Milk, Bread, Butter, Cheese
T4	Bread, Butter, Eggs
T5	Milk, Bread, Butter
T6	Milk, Bread, Juice
T7	Bread, Butter, Cheese
T8	Milk, Eggs

CALCULATION Support(Milk) = 6 / 8 × 100 = 75%.

STEP 1 • L_1

Keep products with support ≥ 50%

Minimum count = 4, so Eggs, Cheese, and Juice cannot continue.

Candidate	Count	Support	Decision
Milk	6	75%	**Keep**
Bread	7	87.5%	**Keep**
Butter	5	62.5%	**Keep**
Eggs	3	37.5%	**Discard**
Cheese	2	25%	**Discard**
Juice	1	12.5%	**Discard**

RESULT L_1 = {Milk}, {Bread}, {Butter}. We now generate pairs only from these three.

Why can we discard Eggs entirely?

If an item occurs in only 3 baskets, any pair containing it occurs in at most 3 baskets.

Eggs

Count = 3 out of 8

Eggs + Milk ≤ 3
Eggs + Bread ≤ 3

Required count = 4

No combination containing Eggs can reach 4.

APRIORI RULE If a smaller itemset is infrequent, every larger itemset containing it is infrequent.

STEP 2 • C_2

Generate candidate pairs from L_1

Only Milk, Bread, and Butter survived the first filtering step.

$$L_1 = \{Milk\} \quad \{Bread\} \quad \{Butter\}$$

Candidate ID	2-itemset (C_2)
P1	Milk + Bread
P2	Milk + Butter
P3	Bread + Butter

NEXT Count how many of the 8 transactions contain BOTH items in each pair.

Does each transaction contain both items?

A tick means the full pair appears in that transaction.

Tx	Milk + Bread	Milk + Butter	Bread + Butter
T1	✓	✓	✓
T2	✓	—	—
T3	✓	✓	✓
T4	—	—	✓
T5	✓	✓	✓
T6	✓	—	—
T7	—	—	✓
T8	—	—	—
TOTAL	5	3	5

Minimum count

4

Pair count

Both present

REFERENCE Source baskets: T1–T8 from the original transaction table.

Keep pairs that appear at least 4 times

Calculate support for each pair out of the same 8 transactions.

Candidate pair	Count	Support calculation	Decision
Milk + Bread	5	5 / 8 = 62.5%	**Keep**
Milk + Butter	3	3 / 8 = 37.5%	**Discard**
Bread + Butter	5	5 / 8 = 62.5%	**Keep**

L_2 = {Milk, Bread} and {Bread, Butter}

NOTE Frequent individual products do NOT guarantee frequent product pairs.

CHECKPOINT

Why did Milk + Butter disappear?

All three products were frequent individually, but one pair failed at 37.5%.

Milk

75%

Butter

62.5%

Milk + Butter

37.5%

Each pair is evaluated using its own transaction count.

NEXT Attempt a 3-product combination using L_2.

STEP 3 • C_3

Try the combination Milk + Bread + Butter

Apriori checks all its 2-item subsets before counting its support.

Possible triple: {Milk, Bread, Butter}

Required 2-item subset	Support (from Step 2)	Frequent?
Milk + Bread	62.5%	**Pass**
Bread + Butter	62.5%	**Pass**
Milk + Butter	37.5%	**Fail**

PRUNE Since Milk + Butter is not frequent, the triple cannot be frequent. Do not count it.

STOPPING CONDITION

No valid 3-item candidates: Apriori stops

The algorithm continues only when it can generate candidates after pruning.



Final frequent itemsets (5)

Milk **Bread** **Butter** **Milk + Bread** **Bread + Butter**

NEXT Use these frequent pairs to generate directional association rules.

Association rules: A → B

A is the antecedent (left side); B is the consequent (right side).

Milk → Bread

Among Milk transactions, how often is Bread present?

Bread → Milk

Among Bread transactions, how often is Milk present?

The pair is the same, but the direction changes the question.

RULES A → B is a purchasing pattern, not evidence that A causes B.

METRIC 1 • DEFINITION

Support = How common is the combination?

Use ALL transactions in the denominator.

Support(Milk, Bread)

5 ÷ 8 = 62.5%

Interpretation

62.5% of all baskets contain both Milk and Bread.

REFERENCE — 8 TRANSACTIONS

ID	Products
T1	Milk, Bread, Butter
T2	Milk, Bread, Eggs
T3	Milk, Bread, Butter, Cheese
T4	Bread, Butter, Eggs
T5	Milk, Bread, Butter
T6	Milk, Bread, Juice
T7	Bread, Butter, Cheese
T8	Milk, Eggs

REMEMBER Support asks: How often are these products purchased together?

METRIC 2 • DEFINITION

Confidence = If A is bought, how often is B?

Unlike Support, confidence uses only transactions containing A.

Rule: Milk → Bread

Confidence(A → B) = Count(A and B) ÷ Count(A)

Count(Milk and Bread) = 5
Count(Milk) = 6

Confidence = 5 ÷ 6 = 83.33%

Interpretation: Of the 6 Milk baskets, 5 also include Bread.

REFERENCE — 8 TRANSACTIONS

ID	Products
T1	Milk, Bread, Butter
T2	Milk, Bread, Eggs
T3	Milk, Bread, Butter, Cheese
T4	Bread, Butter, Eggs
T5	Milk, Bread, Butter
T6	Milk, Bread, Juice
T7	Bread, Butter, Cheese
T8	Milk, Eggs

REMEMBER Confidence is directional: A → B is not necessarily equal to B → A.

Reversing a rule changes Confidence

For Bread → Milk, the denominator is the 7 transactions containing Bread.

REFERENCE — 8 TRANSACTIONS

ID	Products
T1	Milk, Bread, Butter
T2	Milk, Bread, Eggs
T3	Milk, Bread, Butter, Cheese
T4	Bread, Butter, Eggs
T5	Milk, Bread, Butter
T6	Milk, Bread, Juice
T7	Bread, Butter, Cheese
T8	Milk, Eggs

Bread → Milk

Confidence = 5 ÷ 7 = 71.43%

Direction	Confidence
Milk → Bread	83.33%
Bread → Milk	71.43%

KEY IDEA The number of joint purchases stays 5; only the left-hand product count changes.

Evaluate all four rules with min confidence = 80%

Rules below 80% do not pass this chosen threshold.

Association rule	Numerator / denominator	Confidence	Decision
Milk → Bread	5 / 6	83.33%	**Keep**
Bread → Milk	5 / 7	71.43%	**Discard**
Bread → Butter	5 / 7	71.43%	**Discard**
Butter → Bread	5 / 5	100%	**Keep**

Remaining: Milk → Bread and Butter → Bread

QUESTION Both remaining rules have high confidence. Are both equally meaningful?

METRIC 3 • MOTIVATION

Why high Confidence can be misleading

Bread is already common, even without knowing anything about Milk.



Knowing that Milk was bought does not raise Bread’s observed purchase rate.

NEXT Lift compares Confidence with the consequent product’s normal popularity.

METRIC 3 • DEFINITION

Lift = Confidence divided by baseline support

Is the combination actually special?

Lift asks: Does A make B more likely than usual?

Lift(A → B) = Confidence(A → B) ÷ Support(B)

Lift(Milk → Bread) = Confidence(Milk→ Bread) ÷ Support(Bread)

Milk → Bread	Value	Source
Confidence	83.33%	5 / 6
Bread support	87.50%	7 / 8
Lift	0.95	0.8333 / 0.875

HOW TO INTERPRET LIFT

Lift < 1 — **Negative association**

Milk → Bread = 0.95

Lift = 1 — **No association**

A does not change B's rate

Lift > 1 — **Positive association**

Butter → Bread = 1.14

INTERPRET 83.33% Confidence sounds strong, but Bread's baseline is even higher at 87.5%.

Lift for Butter → Bread

Compare the 100% confidence with Bread's baseline support of 87.5%.

Butter → Bread	Value	Source
Both Butter + Bread	5	T1, T3, T4, T5, T7
Butter count	5	T1, T3, T4, T5, T7
Confidence	100%	5 / 5
Bread baseline	87.5%	7 / 8
Lift	1.14	1 ÷ 0.875

LIFT = 1.14

Lift > 1: Positive

Butter buyers purchased Bread at 1.14× the baseline rate.

INTERPRET Positive, but modest, association. The dataset has only eight baskets.

INTERPRETATION

Read Support, Confidence, and Lift together

For the two rules that passed our 80% minimum confidence.

Rule	Support	Confidence	Lift	Interpretation
Milk → Bread	62.5%	83.33%	0.95	High confidence, but no positive association
Butter → Bread	62.5%	100%	1.14	Positive, modest association

Support = prevalence **Confidence = conditional likelihood** **Lift = relative to baseline**

REMEMBER High Confidence alone is not proof of a useful association.

RECAP

Three questions to remember

Teach the algorithm and the evaluation metrics as two connected stages.

SUPPORT	**How often are A and B bought together?**	**5 / 8 = 62.5%**
CONFIDENCE	**If A is bought, how often is B bought?**	**5 / 6 = 83.33%**
LIFT	**Is B more likely than usual when A is bought?**	**83.33% / 87.5% = 0.95**

Apriori uses Support to find itemsets; Confidence and Lift evaluate the rules.